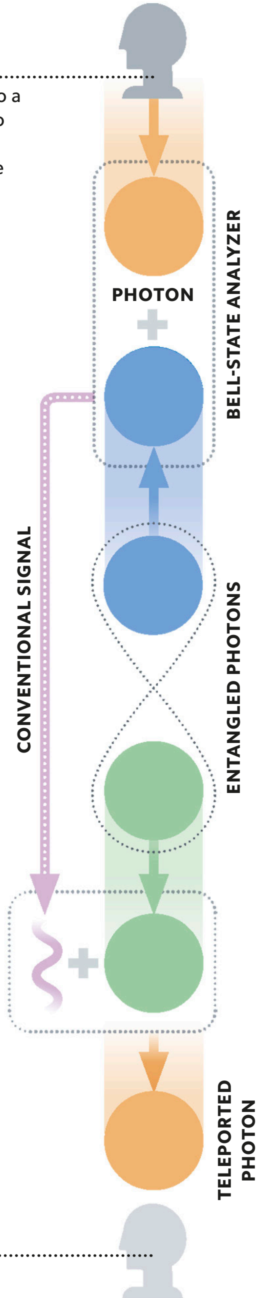
QUANTUM TELEPORTATION

The strange physics of entanglement (see pp.72–73) can be put to work in quantum teleportation—the reconstruction of quantum information from one system in another system. Teleportation involves the creation of an entangled pair of quantum states (qubits; see p.106), which are then separated. The system to be teleported interacts with one qubit, resulting in a "classical" measurement that is then transmitted to the location of the other qubit at the speed of light, and used to reconstruct the interacting system.

SENDER

Alice encodes information into a photon, which is then made to interact with one of a pair of entangled particles in a device called a "Bell-state analyzer."



Bob has the other half of the entangled photon pair. Information about Alice's system, sent by conventional means, allows him to recreate Alice's encoded photon and read its information.

RECEIVER